

a

Biological Model and Silicone Biofilm Substrate

b

Simulated Microgravity and Ground Hardware



Mycobacterium marinum 1218R
BSL-2 surrogate for Mycobacterium tuberculosis
Stable chromosomal RFP at Giles phage attB site

Culture Vessel Specifications (31°C, 4 Days):

- Polydimethylsiloxane (PDMS) silicone membranes
- Suspended biofilm-forming cell pellets harvested for RNA-seq
- $n = 3$ biological replicates per experimental modality

1. Lab-Designed 3D Clinostat ($n = 3$)

Continuous 2-axis clinorotation ($I = 1.5$ rpm, $O = 3.825$ rpm)
Samples: RFP3D11, RFP3D39, RFP3D47

2. Random Positioning Machine 2.0 ($n = 3$)

Random multi-axis velocity vectoring, time-averaged $< 0.01g$
Samples: RFPRPM4, RFPRPM41, RFPRPM6

3. Static 1g Earth Control ($n = 3$)

Stationary incubator shelf adjacent to simulators
Samples: RFPNG14, RFPNG35, RFPNG45

c

Empirical Multi-Scale Analytical Framework

